Does the connected Wuji Hand 2 have tactile sensors?
这只手有没有触觉传感器？

Device: Wuji Hand 2.2 (Beta), right hand, serial WH2KA01260824017, hw 0.2.0 · firmware v2.5.1 → v3.0.0 (upgraded 2026-11-13) · wuji_sdk 2026.11.9 · Ethernet/UDP 192.168.1.111 · tests run 2026-11-13, 18:38–20:04 EST

Verdict: NO — no usable tactile sensor (无可用的触觉传感器). The hand's controller never finds fingertip tactile electronics on any of the five finger buses, so no tactile data can be read. The fingertips do carry sensor-style pads with a flex circuit (柔性电路板, FPC), so this unit is either the non-tactile build of the 2.2 Beta or has its fingertip electronics disconnected on all five fingers — from the computer's side both mean the same: no tactile sensing.

How it was identified (鉴定方法)

The Wuji Hand 2 puts each fingertip tactile module on the finger's internal bus as node 5 (nodes 1–4 are the joints). Six independent checks were read through the official SDK and Wuji CLI, repeated with motors off and on, after a software reboot, and again after flashing the current firmware v3.0.0 — the hand's own bookkeeping, not a guess from the data stream.

Check (检查项)	Expected if tactile modules exist	Observed — v2.5.1 and v3.0.0, all identical
node_online bitmask (bus enumeration, 总线枚举)	bit of node 5 set on each finger	0xF7BDEF: joints 1–4 online on all 5 fingers, node 5 absent on all 5
slave_version table	25 entries (20 joints + 5 tactile)	20 entries — joints only
fingertip_online() / comm tactile_online_mask	True / bits set	[False]×5 / 0
tactile_timeout_total (polls that failed)	> 0 if a module were fitted but dead	[0,0,0,0,0] — the master never polls: nothing was enumerated at boot
fingertip_<f>_status / info	model + state, 40/34-point descriptor	v2.5.1: InternalError; v3.0.0: NodeOfflineError “node not on the bus; check that the fingertip module is connected”
fingertip/<f>/data streams, 3 s, pads pressed	~100 Hz frames	0 frames on all five
Joints (control check, 对照)	20/20 online	20/20 online; playback lap mean joint error 1.85° — the hand itself is healthy

Why the answer is “no” (理由)

1. The electronics are absent from the bus, not merely silent. A fitted-but-faulty module makes the controller poll it and time out; here the timeout counters stay at 0 and node 5 is never listed. All five fingers show the same, twice (before/after reboot) and on two firmware versions.

2. Software is ruled out. Firmware v3.0.0 pairs with the installed SDK 2026.11.9 (Wuji's current pairing); the tactile API requires only firmware ≥ v2.1.0 and SDK ≥ 2026.7.21. Motors enabled vs. disabled made no difference. Wuji CLI's doctor finds the hand but has no tactile verdict for Hand 2.

3. The hardware is consistent with the non-tactile 2.2 Beta build. Wuji's current Hand 2.2 (Beta) manual is explicitly the “Non-Tactile” version and that variant has fingertip soft pads; the tactile variant adds a 40-point (thumb) / 34-point (other fingers) 3-axis force array per fingertip at 100 Hz. The pads on this unit carry a flex circuit (photo), so sensor pads may be mounted — but no fingertip electronics answer. Only Wuji can say whether this is a non-tactile assembly or a disconnected one; both give no tactile data.



Thumb fingertip with the silicone pad pulled back: metal core with an orange flex circuit — sensor-style hardware, yet node 5 never appears on the bus.

Evidence files (runs/tactile/ in the wuji_hand_poliocy repo): probe_2026-11-13_184739.txt · reboot_probe_2026-11-13_192609.txt · wuji_cli_2026-11-13_195354.txt · wuji_upgrade_2026-11-13_200001.txt · WUJI_SUPPORT_SUMMARY.md

What would change the verdict: fingertip_online() returning True for any finger, or node 5 appearing in node_online. The live viewer (deploy/tactile_viewer.py) re-checks every 2 s and will show the data at once.